■ Supplementary Material

For the manuscript "Smart Predict-Then-Optimize Based Unit Commitment for Integrated Energy Systems".

This material, as a supplement to the manuscript "*Smart Predict-Then-Optimize Based Unit Commitment for Integrated Energy Systems*", provides additional information not presented in the manuscript due to page limitations. Appendix A provides the detailed model of the TPO-UC model for HE-IES. For the convenience of readers, the nomenclature is presented first, followed by a detailed introduction to the TPO-UC model. Appendix B provides the proof for *Proposition* 2.

APPENDIX A
DETAILED MODEL OF THE TPO-UC MODEL FOR HE-IES

■ *Nomenclature*

1) Indices and Sets

i	Index for units (thermal, CHP, RES, etc.)
l	Index for time periods or operational states.
t	Time period index.
T	Set of time periods.
$I_m, I_{chp}, I_{res}, I_{load}$	Sets of thermal units, CHP units, renewable energy sources, and loads.
E^i	Set of the feasible domain vertices for CHP unit i.
K_n, K_L	Intermediate and load nodes in the heating network.

2) Variables

$p_{tu}^{i,t}$	Power output of thermal unit i at time t.
$p_{chp}^{i,t}$	Power output of CHP unit i at time t.
$p_{res}^{i,t}$	Power output of RES i at time t.
$p_{LS}^{i,t}$	Load shedding at node i, time t.
$u_{tu}^{i,t}, v_{tu}^{i,t}, w_{tu}^{i,t}$	Binary variables for unit commitment (on/off, startup, shutdown).
$\xi_{chp}^{i,t,k}$	The k_{th} Fraction of CHP unit i at time t.
$\tau_{s,out}^{j,i,t}$	Outlet temperature of supply pipe j at node i, time t.
$\tau_{building}^{i,t}$	Indoor temperature of building i at time t.

3) Parameters

$C_{tu}^{2,l}, C_{tu}^{1,l}, C_{tu}^{0,l}$	Quadratic, linear, and fixed cost coefficients for thermal units.
$\underline{P}_m^i, \overline{P}_m^i$	Minimum/maximum power output of thermal unit i.
RU_{tu}^i, RD_{tu}^i	Ramp-up/ramp-down limits of thermal unit i.
SRU^i, SRD^i	System-wide spinning reserve requirements.
λ_p^j	Heat loss coefficient of pipe j.
τ_{amb}	Ambient temperature.
c_w	Specific heat capacity of water.

■ ***TPO-UC model for HE-IES***

The TPO-UC model for HE-IES can be stated as a two-stage problem, as shown in optimization problem below. The first stage involves day-ahead UC decisions (here-and-now decisions), while the second stage addresses real-time re-dispatch adjustments to account for deviations in renewable energy (wait-and-see decisions).

4) Day-ahead UC model

$$\min_{\substack{p_{tu}^{i,t},u_{tu}^{i,t},v_{tu}^{i,t},w_{tu}^{i,t} \\ i\in(I_t\cup I_{chp}\cup I_{res}\cup I_{load})}} \sum_{t\in T}\left(\sum_{i\in I_t} C_{tu}^{i,t}+\sum_{i\in I_{chp}} C_{chp}^{i,t}+\sum_{i\in I_{res}} C_{res}^{i,t}+\sum_{i\in I_{load}} C_{load}^{i,t}\right), \tag{A.1}$$

wherein

$$C_{tu}^{i,t}=C_{tu}^{1,i}p_{tu}^{i,t}+C_{tu}^{0,i}u_{tu}^{i,t}+C_{tu}^{U,i}v_{tu}^{i,t}+C_{tu}^{D,i}w_{tu}^{i,t}, \tag{A.2}$$

$$C_{chp}^{i,t}=C_{chp}^{1,i}p_{chp}^{i,t}+C_{chp}^{0,i}u_{chp}^{i,t}+C_{chp}^{U,i}v_{chp}^{i,t}+C_{chp}^{D,i}w_{chp}^{i,t}+C_{chp}^{1,i}h_{chp}^{i,t}, \tag{A.3}$$

$$C_{res}^{i,t}=C_{res}^{i}(\overline{P}_{res}^{i,t}-p_{res}^{i,t}), \tag{A.4}$$

$$C_{load}^{i,t}=C_{load}^{i}p_{LS}^{i,t}, \tag{A.5}$$

$$\text{s.t.}\qquad u_{tu}^{i,t}\underline{P}_{tu}^{i}\le p_{tu}^{i,t}\le u_{tu}^{i,t}\overline{P}_{tu}^{i},\quad \forall i\in I_{tu},t\in T, \tag{A.6}$$

$$0\le p_{res}^{i,t}\le \hat{p}_{res}^{i,t},\quad \forall i\in I_{res},t\in T, \tag{A.7}$$

$$0\le p_{LS}^{i,t}\le P_{load}, \tag{A.8}$$

$$\begin{cases} p_{chp}^{i,t}=\sum_{k\in E^i}\xi_{chp}^{i,t,k}P_{chp}^{i,k} & ,\forall i\in I_{chp},t\in T \\ h_{chp}^{i,t}=\sum_{k\in E^i}\xi_{chp}^{i,t,k}H_{chp}^{i,k} & ,\forall i\in I_{chp},t\in T \\ 0\le \xi_{chp}^{i,t,k}\le 1 & ,\forall i\in I_{chp},t\in T,k\in E^i \\ \sum_{k\in E^i}\xi_{chp}^{i,t,k}=1 & ,\forall i\in I_{chp},t\in T \end{cases}, \tag{A.9}$$

$$-RD_{tu}^{i}\Delta t\le p_{tu}^{i,t}-p_{tu}^{i,t-1}\le RU_{tu}^{i}\Delta t,\forall i\in I_{tu},t\in T, \tag{A.10}$$

$$-RD_{chp}^{i}\Delta t\le p_{chp}^{i,t}-p_{chp}^{i,t-1}\le RU_{chp}^{i}\Delta t,\forall i\in I_{chp},t\in T, \tag{A.11}$$

$$u_{tu}^{i,t}-u_{tu}^{i,t-1}=v_{tu}^{i,t}-w_{tu}^{i,t},\forall i\in I_{tu},t\in T, \tag{A.12}$$

$$\sum_{y=t-T_{tu}^{+,i}+1}^{t} v_{tu}^{i,y}\le u_{tu}^{i,t},\forall i\in I_{tu},t\in T,\forall i\in I_{tu},t\in T, \tag{A.13}$$

$$\sum_{y=t-T_{tu}^{-,i}+1}^{t} w_{tu}^{i,y}\le 1-u_{tu}^{i,t},\forall i\in I_{tu},t\in T, \tag{A.14}$$

$$0\le p_{ru}^{i,t}\le \min\left\{u_{tu}^{i,t}RU_{tu}^{i}\Delta t,u_{tu}^{i,t}\overline{P}_{tu}^{i}-p_{tu}^{i,t}\right\},\forall i\in I_{tu},t\in T, \tag{A.15}$$

$$0 \le p_{rd}^{i,t} \le \min\left\{u_{tu}^{i,t} RD_{tu}^{i} \Delta t,\ p_{tu}^{i,t} - u_{tu}^{i,t} \overline{P}_{tu}^{i}\right\}, \forall i \in I_{tu}, t \in T\ , \tag{A.16}$$

$$\sum_{i \in I_{tu}} p_{ru}^{i,t} \ge SRU^{t}, \forall t \in T\ , \tag{A.17}$$

$$\sum_{i \in I_{tu}} p_{rd}^{i,t} \ge SRD^{t}, \forall t \in T\ , \tag{A.18}$$

$$\sum_{i \in I_{tu}} p_{tu}^{i,t} + \sum_{i \in I_{chp}} p_{chp}^{i,t} + \sum_{i \in I_{res}} p_{res}^{i,t} = \sum_{i \in I_{load}} (P_{load}^{i,t} - p_{LS}^{i,t}), \forall t \in T\ , \tag{A.19}$$

$$\underline{P}^{b} \le \sum_{i \in I_{tu}} f_{tu}^{i,b} p_{tu}^{i,t} + \sum_{i \in I_{chp}} f_{chp}^{i,b} p_{chp}^{i,t} + \sum_{i \in I_{res}} f_{res}^{i,b} p_{res}^{i,t} - \sum_{i \in I_{load}} f_{load}^{i,b} (P_{load}^{i,t} - p_{LS}^{i,t}) \le \overline{P}^{b}\ , \tag{A.20}$$

$$\tau_{s,out}^{j,t,i} = (1-\eta_{p}^{j,i})((1-\alpha_{p}^{j,i})\tau_{s,in}^{j,t-\gamma_{p}^{j,i}-1,i} + \alpha_{p}^{j,i}\tau_{s,in}^{j,t-\gamma_{p}^{j,i},i}) + \eta_{p}^{j,i}\tau_{amb}, \forall j \in \Phi_{p}, t \in T_{h}, i \in I_{chp}\ , \tag{A.21}$$

$$\tau_{r,out}^{j,t,i} = (1-\eta_{p}^{j,i})((1-\alpha_{p}^{j,i})\tau_{r,in}^{j,t-\gamma_{p}^{j,i}-1,i} + \alpha_{p}^{j,i}\tau_{r,in}^{j,t-\gamma_{p}^{j,i},i}) + \eta_{p}^{j,i}\tau_{amb}, \forall j \in \Phi_{p}, t \in T_{h}, i \in I_{chp}\ , \tag{A.22}$$

$$\begin{cases} \gamma_{p}^{j,i} = \left\lceil \dfrac{\rho_{w} A_{p}^{j} L_{p}^{j}}{m_{p}^{j} \Delta t_{h}} \right\rceil - 1 \\ R_{p}^{j,i} = (\gamma_{p}^{j} + 1) m_{p}^{j} \Delta t_{h} \\ \alpha_{p}^{j,i} = \dfrac{R_{p}^{j} - \rho_{w} A_{p}^{j} L_{p}^{j}}{m_{p}^{j} \Delta t_{h}} \\ \eta_{p}^{j,i} = 1 - e^{-\frac{\lambda_{p}^{j} \Delta t_{h}}{\rho_{w} A_{p}^{j} c_{w}} (\gamma_{p}^{j} + 1.5 - \alpha_{p}^{j})} \end{cases}\ , \tag{A.23}$$

$$\tau_{s,n}^{1,t,i} = \tau_{s,in}^{1,t,i}\,|_{j \in \Phi_{from}}, \forall t \in T_{h}, i \in I_{chp}\ , \tag{A.24}$$

$$\tau_{r,n}^{1,t,i} = \tau_{r,out}^{1,t,i}\,|_{j \in \Phi_{from}}, \forall t \in T_{h}, i \in I_{chp}\ , \tag{A.25}$$

$$h_{chp}^{l,t,i} = c_{w} m_{p}^{1,i} (\tau_{s,n}^{1,t,i} - \tau_{r,n}^{1,t,i}), \forall t \in T_{h}, i \in I_{chp}\ , \tag{A.26}$$

$$\sum_{j \in \Phi_{p\,from}^{k}} m_{p}^{j,i} \tau_{s,n}^{k,t,i} = m_{p}^{j,i} \tau_{s,out}^{j,t,i}\,|_{j \in \Phi_{Pto}^{k}}, \forall t \in T_{h}, k \in K_{n}, i \in I_{chp}\ , \tag{A.27}$$

$$m_{p}^{j,i} \tau_{r,n}^{k,t,i}\,|_{j \in \Phi_{Pto}^{k}} = \sum_{j \in \Phi_{p\,from}^{k}} m_{p}^{j,i} \tau_{r,out}^{j,t,i}, \forall t \in T_{h}, k \in K_{n}, i \in I_{chp}\ , \tag{A.28}$$

$$\tau_{s,n}^{k,t,i} = \tau_{s,in}^{j,t,i}, \forall j \in \Phi_{p_{from}}^{k}, t \in T_{h}, k \in K_{n}, i \in I_{chp}\ , \tag{A.29}$$

$$\tau_{r,n}^{k,t,i} = \tau_{r,in}^{j,t,i}, \forall j \in \Phi_{p_{to}}^{k}, t \in T_{h}, k \in K_{n}, i \in I_{chp}\ , \tag{A.30}$$

$$\tau_{s,n}^{k,t,i} = \tau_{s,out}^{j,t,i}, \forall j \in \Phi_{p_{to}^{k}}, k \in K_{L}, t \in T_{h}, i \in I_{chp}\ , \tag{A.31}$$

$$\tau_{r,n}^{k,t,i} = \tau_{r,in}^{j,t,i}, \forall j \in \Phi_{p_{to}^{k}}, k \in K_{L}, t \in T_{h}, i \in I_{chp}\ , \tag{A.32}$$

$$h_{chp}^{l,t,i} = c_w m_p^{j,i}(\tau_{s,n}^{k,t,i} - \tau_{r,n}^{k,t,i}), \forall j \in \Phi_{p_{to}}^{k}, k \in K_L, t \in T_h, i \in I_{chp}, l \in L, \tag{A.33}$$

$$\underline{\tau}_{s,n}^{k,i} \le \tau_{s,n}^{k,t,i} \le \overline{\tau}_{s,n}^{k,i}, \forall k \in \Phi_n, t \in T_h, i \in I_{chp}, \tag{A.34}$$

$$\underline{\tau}_{r,n}^{k,i} \le \tau_{r,n}^{k,t,i} \le \overline{\tau}_{r,n}^{k,i}, \forall k \in \Phi_n, t \in T_h, i \in I_{chp}, \tag{A.35}$$

$$\lambda^{l,i} = e^{-\Delta t_h / R_s^{l,i} C_{air}^{l,i}}, \tag{A.36}$$

$$\tau_{building}^{l,t,i} = \lambda^{l,i} \tau_{building}^{l,t-1,i} + (1-\lambda^{l,i})(\tau_{outside}^{t,i} + \frac{R_s^{l,i}}{N^{l,i}} h_{load}^{l,t,i}), \forall l \in L, t \in T_h, i \in I_{chp}, \tag{A.37}$$

$$\underline{\tau}_{building} \le \tau_{building}^{l,t,i} \le \overline{\tau}_{building}, \forall l \in L, t \in T_h, i \in I_{chp}, \tag{A.38}$$

$$\tau_{building}^{l,end,i} = \tau_{opt}, \forall l \in L, i \in I_{chp}, \tag{A.39}$$

$$\sum_{t \in T_h} \tau_{building}^{l,t,i} = N_t \tau_{opt}, \forall l \in L, i \in I_{chp}, \tag{A.40}$$

$$\sum_j \frac{\tau_{s,n}^{k,end,i} + \tau_{s,n}^{k',end,i}}{2} m_p^{j,i} \ge \sum_j \frac{\tau_{s,n}^{k,l,i} + \tau_{s,n}^{k',end,i}}{2} m_p^{j,i}, \forall k \in K, i \in I_{chp}, \tag{A.41}$$

$$\sum_j \frac{\tau_{r,n}^{k,end,i} + \tau_{r,n}^{k',end,i}}{2} m_p^{j,i} \ge \sum_j \frac{\tau_{r,n}^{k,l,i} + \tau_{r,n}^{k',end,i}}{2} m_p^{j,i}, \forall k \in K, i \in I_{chp}, \tag{A.42}$$

$$h_{chp}^{i,t} \Delta t = \sum_{t_h = \frac{\Delta t}{\Delta t_h}(t-1)+1}^{\frac{\Delta t}{\Delta t_h} t} h_{chp}^{i,t_h} \Delta t_h, \forall t \in T. \tag{A.43}$$

The objective function (A.1) minimizes the total cost of operating the integrated energy system, including thermal generation costs ($C_{tu}^{i,l}$), CHP operational costs ($C_{chp}^{i,l}$), RES curtailment penalties ($C_{res}^{i,l}$), and load shedding penalties ($C_{load}^{i,l}$). Equations (A.2)-(A.5) define the cost functions for thermal units, CHP units, renewable energy sources, and load shedding, respectively, incorporating linear, and fixed cost components. Power output limits for thermal units, renewable sources, and load shedding are enforced through constraints (A.6)-(A.8), ensuring generation remains within technical bounds. Constraint (A.9) represents the CHP coupling constraints. It models the CHP operation using a convex combination of feasible domain vertices, enforcing the coupling between the electricity and heat subsystems. Constraints (A.10)-(A.11) limit the ramp rate for thermal and CHP units. Constraints (A.12)-(A.14) represents unit commitment logic, including startup and shutdown dynamics. System reliability is ensured via spinning reserve requirements (A.15)-(A.18), which mandate sufficient upward and downward reserves to handle contingencies. Constraint (A.19) guarantees the power balance between generation and demand, while constraints (A.20) enforces transmission line flow limits.

The constraints of the heat network include (A.21)-(A.43). The heating network quasi-dynamics are captured by constraints (A.21)-(A.23), which model temperature propagation in supply and return pipes, accounting for heat loss and time delays. Constraints (A.24)-(A.33) enforce energy balance at source, intermediate, and load nodes, ensuring consistent mass and heat flow. Temperature limits (A.34)-(A.35) and thermal comfort constraints (A.36)-(A.40) regulate indoor building temperatures, maintaining occupant comfort while optimizing energy use. Constraints (A.41) -(A.42) specify the initial and final state conditions for the heating network. Constraint (A.43) address the coupling

between electricity and heat systems.

5) Re-dispatch model

$$\min_{\substack{p_{tu}^{i,t,+},p_{tu}^{i,t,-},p_{chp}^{i,t,+},p_{chp}^{i,t,-}\\ h_{chp}^{i,t,+},h_{chp}^{i,t,-}}} \sum_{t\in T}\sum_{i\in I}(\gamma_{tu}^{i,+}p_{tu}^{i,t,+}+\gamma_{tu}^{i,-}p_{tu}^{i,t,-}+\gamma_{chp}^{i,+}p_{chp}^{i,t,+}+\gamma_{chp}^{i,-}p_{chp}^{i,t,-}+\gamma_{hchp}^{i,+}h_{chp}^{i,t,+}+\gamma_{hchp}^{i,-}h_{chp}^{i,t,-}) \text{ ,(A.44)}$$

$$\text{s.t.} \quad p_{tu}^{i,t}+p_{tu}^{i,t+}\le u_{tu}^{i,t}\overline{P}_{tu}^{i}, \quad \forall i\in I_{tu},t\in T \text{ ,} \tag{A.45}$$

$$p_{tu}^{i,t}-p_{tu}^{i,t-}\ge u_{tu}^{i,t}\underline{P}_{tu}^{i}, \quad \forall i\in I_{tu},t\in T \text{ ,} \tag{A.46}$$

$$0\le p_{res_{stage2}}^{i,t}\le \tilde{p}_{res}^{i,t}, \quad \forall i\in I_{res},t\in T \text{ ,} \tag{A.47}$$

$$\begin{cases} p_{chp}^{i,t}+p_{chp}^{i,t+}-p_{chp}^{i,t-}=\sum_{k\in E^i}\xi_{chp_{stage2}}^{i,t,k}P_{chp}^{i,k} & ,\forall i\in I_{chp},t\in T\\ h_{chp}^{i,t}+h_{chp}^{i,t+}-h_{chp}^{i,t-}=\sum_{k\in E^i}\xi_{chp_{stage2}}^{i,t,k}H_{chp}^{i,k} & ,\forall i\in I_{chp},t\in T\\ 0\le \xi_{chp_{stage2}}^{i,t,k}\le 1 & ,\forall i\in I_{chp},t\in T,k\in E^i\\ \sum_{k\in E^i}\xi_{chp_{stage2}}^{i,t,k}=1 & ,\forall i\in I_{chp},t\in T \end{cases} , \tag{A.48}$$

$$-RD_{tu}^{i}\Delta t\le (p_{tu}^{i,t}+p_{tu}^{i,t+}-p_{tu}^{i,t-})-(p_{tu}^{i,t-1}+p_{tu}^{i,(t-1)+}-p_{tu}^{i,(t-1)-})\le RU_{tu}^{i}\Delta t,\forall i\in I_{tu},t\in T \text{ ,} \tag{A.49}$$

$$-RD_{chp}^{i}\Delta t\le (p_{chp}^{i,t}+p_{chp}^{i,t+}-p_{chp}^{i,t-})-(p_{chp}^{i,t-1}+p_{chp}^{i,(t-1)+}-p_{chp}^{i,(t-1)-})\le RU_{chp}^{i}\Delta t,\forall i\in I_{chp},t\in T \text{ ,(A.50)}$$

$$0\le p_{ru_{stage2}}^{i,t}\le \min\left\{u_{tu}^{i,t}RU_{tu}^{i}\Delta t,u_{tu}^{i,t}\overline{P}_{tu}^{i}-(p_{tu}^{i,t}+p_{tu}^{i,t+}-p_{tu}^{i,t-})\right\},\forall i\in I_{tu},t\in T \text{ ,} \tag{A.51}$$

$$0\le p_{rd_{stage2}}^{i,t}\le \min\left\{u_{tu}^{i,t}RD_{tu}^{i}\Delta t,(p_{tu}^{i,t}+p_{tu}^{i,t+}-p_{tu}^{i,t-})-u_{tu}^{i,t}\overline{P}_{tu}^{i}\right\},\forall i\in I_{tu},t\in T \text{ ,} \tag{A.52}$$

$$\sum_{i\in I_{tu}}p_{ru_{stage2}}^{i,t}\ge SRU^{t},\forall t\in T \text{ ,} \tag{A.53}$$

$$\sum_{i\in I_{tu}}p_{rd_{stage2}}^{i,t}\ge SRD^{t},\forall t\in T \text{ ,} \tag{A.54}$$

$$\sum_{i\in I_{tu}}(p_{tu}^{i,t}+p_{tu}^{i,t+}-p_{tu}^{i,t-})+\sum_{i\in I_{chp}}(p_{chp}^{i,t}+p_{chp}^{i,t+}-p_{chp}^{i,t-})+\sum_{i\in I_{res}}p_{res_{stage2}}^{i,t}=\sum_{i\in I_{load}}(P_{load}^{i,t}-p_{LS_{stage2}}^{i,t}),\forall t\in T \text{ ,}$$

(A.55)

$$\underline{P}^{b}\le \sum_{i\in I_{tu}}f_{tu}^{i,b}(p_{tu}^{i,t}+p_{tu}^{i,t+}-p_{tu}^{i,t-})+\sum_{i\in I_{chp}}f_{chp}^{i,b}(p_{chp}^{i,t}+p_{chp}^{i,t+}-p_{chp}^{i,t-})+\sum_{i\in I_{res}}f_{res}^{i,b}p_{res_{stage2}}^{i,t}-\sum_{i\in I_{load}}f_{load}^{i,b}(P_{load}^{i,t}-p_{LS_{stage2}}^{i,t})\le \overline{P}^{b} \text{ ,}$$

(A.56)

$$\tau_{s,out_{stage2}}^{j,t,i}=(1-\eta_p^{j,i})((1-\alpha_p^{j,i})\tau_{s,in_{stage2}}^{j,t-\gamma_p^{j,i}-1,i}+\alpha_p^{j,i}\tau_{s,in_{stage2}}^{j,t-\gamma_p^{j,i},i})+\eta_p^{j,i}\tau_{amb},\forall j\in\Phi_p,t\in T_h,i\in I_{chp} \text{ ,(A.57)}$$

$$\tau_{r,out_{stage2}}^{j,t,i}=(1-\eta_p^{j,i})((1-\alpha_p^{j,i})\tau_{r,in_{stage2}}^{j,t-\gamma_p^{j,i}-1,i}+\alpha_p^{j,i}\tau_{r,in_{stage2}}^{j,t-\gamma_p^{j,i},i})+\eta_p^{j,i}\tau_{amb},\forall j\in\Phi_p,t\in T_h,i\in I_{chp} \text{ ,(A.58)}$$

$$\begin{cases} \gamma_p^{j,i} = \left\lceil \dfrac{\rho_w A_p^j L_p^j}{m_p^j \Delta t_h} \right\rceil - 1 \\ R_p^{j,i} = (\gamma_p^j + 1) m_p^j \Delta t_h \\ \alpha_p^{j,i} = \dfrac{R_p^j - \rho_w A_p^j L_p^j}{m_p^j \Delta t_h} \\ \eta_p^{j,i} = 1 - e^{-\frac{\lambda_p^j \Delta t_h}{\rho_w A_p^j c_w}(\gamma_p^j + 1.5 - \alpha_p^j)} \end{cases}, \tag{A.59}$$

$$\tau_{s,n_{stage2}}^{1,t,i} = \tau_{s,in_{stage2}}^{1,t,i} \mid_{j \in \Phi_{from}}, \forall t \in T_h, i \in I_{chp}, \tag{A.60}$$

$$\tau_{r,n_{stage2}}^{1,t,i} = \tau_{r,out_{stage2}}^{1,t,i} \mid_{j \in \Phi_{from}}, \forall t \in T_h, i \in I_{chp}, \tag{A.61}$$

$$h_{chp_{stage2}}^{l,t,i} = c_w m_p^{1,i} (\tau_{s,n_{stage2}}^{1,t,i} - \tau_{r,n_{stage2}}^{1,t,i}), \forall t \in T_h, i \in I_{chp}, \tag{A.62}$$

$$\sum_{j \in \Phi_{p_{from}}^k} m_p^{j,i} \tau_{s,n_{stage2}}^{k,t,i} = m_p^{j,i} \tau_{s,out_{stage2}}^{j,t,i} \mid_{j \in \Phi_{p_{to}}^k}, \forall t \in T_h, k \in K_n, i \in I_{chp}, \tag{A.63}$$

$$m_p^{j,i} \tau_{r,n_{stage2}}^{k,t,i} \mid_{j \in \Phi_{p_{to}}^k} = \sum_{j \in \Phi_{p_{from}}^k} m_p^{j,i} \tau_{r,out_{stage2}}^{j,t,i}, \forall t \in T_h, k \in K_n, i \in I_{chp}, \tag{A.64}$$

$$\tau_{s,n_{stage2}}^{k,t,i} = \tau_{s,in_{stage2}}^{j,t,i}, \forall j \in \Phi_{p_{from}}^k, t \in T_h, k \in K_n, i \in I_{chp}, \tag{A.65}$$

$$\tau_{r,n_{stage2}}^{k,t,i} = \tau_{r,in_{stage2}}^{j,t,i}, \forall j \in \Phi_{p_{to}}^k, t \in T_h, k \in K_n, i \in I_{chp}, \tag{A.66}$$

$$\tau_{s,n_{stage2}}^{k,t,i} = \tau_{s,out_{stage2}}^{j,t,i}, \forall j \in \Phi_{p_{to}^k}, k \in K_L, t \in T_h, i \in I_{chp}, \tag{A.67}$$

$$\tau_{r,n_{stage2}}^{k,t,i} = \tau_{r,in_{stage2}}^{j,t,i}, \forall j \in \Phi_{p_{to}^k}, k \in K_L, t \in T_h, i \in I_{chp}, \tag{A.68}$$

$$h_{chp_{stage2}}^{l,t,i} = c_w m_p^{j,i} (\tau_{s,n_{stage2}}^{k,t,i} - \tau_{r,n_{stage2}}^{k,t,i}), \forall j \in \Phi_{p_{to}}^k, k \in K_L, t \in T_h, i \in I_{chp}, l \in L, \tag{A.69}$$

$$\underline{\tau}_{s,n}^{k,i} \le \tau_{s,n_{stage2}}^{k,t,i} \le \overline{\tau}_{s,n}^{k,i}, \forall k \in \Phi_n, t \in T_h, i \in I_{chp}, \tag{A.70}$$

$$\underline{\tau}_{r,n}^{k,i} \le \tau_{r,n_{stage2}}^{k,t,i} \le \overline{\tau}_{r,n}^{k,i}, \forall k \in \Phi_n, t \in T_h, i \in I_{chp}, \tag{A.71}$$

$$\lambda^{l,i} = e^{-\Delta t_h / R_s^{l,i} C_{air}^{l,i}}, \tag{A.72}$$

$$\tau_{building_{stage2}}^{l,t,i} = \lambda^{l,i} \tau_{building_{stage2}}^{l,t-1,i} + (1 - \lambda^{l,i})(\tau_{outside}^{t,i} + \frac{R_s^{l,i}}{N^{l,i}} h_{load_{stage2}}^{l,t,i}), \forall l \in L, t \in T_h, i \in I_{chp}, \tag{A.73}$$

$$\underline{\tau}_{building} \le \tau_{building_{stage2}}^{l,t,i} \le \overline{\tau}_{building}, \forall l \in L, t \in T_h, i \in I_{chp}, \tag{A.74}$$

$$\tau_{building_{stage2}}^{l,end,i} = \tau_{opt}, \forall l \in L, i \in I_{chp}, \tag{A.75}$$

$$\sum_{t \in T_h} \tau_{building_{stage2}}^{l,t,i} = N_t \tau_{opt}, \forall l \in L, i \in I_{chp}, \tag{A.76}$$

$$\sum_{j} \frac{\tau_{s,n_{stage2}}^{k,end,i} + \tau_{s,n_{stage2}}^{k',end,i}}{2} m_p^{j,i} \geq \sum_{j} \frac{\tau_{s,n_{stage2}}^{k,l,i} + \tau_{s,n_{stage2}}^{k',end,i}}{2} m_p^{j,i}, \forall k \in K, i \in I_{chp}, \tag{A.77}$$

$$\sum_{j} \frac{\tau_{r,n_{stage2}}^{k,end,i} + \tau_{r,n_{stage2}}^{k',end,i}}{2} m_p^{j,i} \geq \sum_{j} \frac{\tau_{r,n_{stage2}}^{k,l,i} + \tau_{r,n_{stage2}}^{k',end,i}}{2} m_p^{j,i}, \forall k \in K, i \in I_{chp}, \tag{A.78}$$

$$h_{chp_{stage2}}^{i,t} \Delta t = \sum_{t_h = \frac{\Delta t}{\Delta t_h}(t-1)+1}^{\frac{\Delta t}{\Delta t_h} t} h'^{i,t_h}_{chp_{stage2}} \Delta t_h, \forall t \in T. \tag{A.79}$$

Objective function (A.44) minimizes the total re-dispatch cost, penalizing upward/downward adjustments in thermal and CHP power outputs ($p_{tu}^{i,t+}, p_{tu}^{i,t-}, p_{chp}^{i,t+}, p_{chp}^{i,t-}$) and heat outputs ($h_{chp}^{i,t+}, h_{chp}^{i,t-}$), weighted by their respective cost coefficients ($\gamma_{tu}^{+}, \gamma_{tu}^{-}, \gamma_{chp}^{+}, \gamma_{chp}^{-}, \gamma_{hchp}^{+}, \gamma_{hchp}^{-}$).

Constraints (A.45)–(A.47) enforce power output limits for adjusted thermal units and renewable sources, where (A.47) ensures RES generation does not exceed real-time realized values ($\tilde{p}_{res}^{i,t}$). Constraint (A.48) represents the CHP coupling constraints in the re-dispatch stage. Ramp rate constraints (A.49)–(A.50) restrict the rate of adjusted thermal and CHP outputs. Spinning reserve requirements (A.51)–(A.54) ensure sufficient flexibility for real-time contingencies, while (A.55) enforces real-time power balance, integrating deviations from the day-ahead schedule. The constraints of the heat network include (A.57)-(A.79), which are almost the same as the heat network constraints in the first stage. The only difference is that all the variables in the heat network have the subscript "stage2", reflecting the actual values of the heat network variables in the re-dispatching stage.

APPENDIX B
PROOF FOR PROPOSITION 2

We establish the proof through the following referenced Theorems and lemmas. The MA-R&D algorithm consists of three nested loops:

1) **Outer loop (MA-R&D):** Computes the whole BiMILP by the reformulation and decomposition strategy. The corresponding counting variable is l.

2) **Middle loop ($\boldsymbol{\rho}$ search):** Adjusts the penalty parameter ρ. The corresponding counting variable is j.

3) **Inner loop ($\boldsymbol{l_1}$-PADM with $\boldsymbol{\rho}$ fixed):** Solves the penalized problem MP^{M} using ADM. The corresponding counting variable is k.

The following convergence result for the l_1-PADM method is adapted from [1].

Theorem B.1 [1]**.** Consider the inner loop of l_1-PADM applied to the penalized strong-duality-based single-level reformulation of the bilevel problem (with a linear follower problem) for a fixed $\rho > 0$. Let $\left\{\left(m^{(j,k)}, n^{(j,k)}\right)\right\}_{k=0}^{\infty}$ be the generated sequence of iterates. If the objective function of the bilevel problem is continuously differentiable and one of the two subproblems in l_1-PADM always has a unique solution, then each convergent subsequence of $\left\{\left(m^{(j,k)}, n^{(j,k)}\right)\right\}_{k=0}^{\infty}$ converges to a stationary point of the penalized strong-duality-based single-level reformulation.

Theorem B.2 [1]**.** Suppose that $\rho^{(j)} \to \infty$, and let $\left\{\left(m^{(j)}, n^{(j)}\right)\right\}_{j=0}^{\infty}$ be a sequence of stationary points of the penalized strong-duality-based single-level reformulation (with fixed $\rho = \rho^{(j)}$) with $\left(m^{(j)}, n^{(j)}\right) \to (m^*, n^*)$ for $j \to \infty$. If, in addition, (m^*, n^*) is bilevel feasible (the value of penalty term equals to zero), then (m^*, n^*) is a stationary point of the strong-duality-based single-level reformulation.

Remark B.1. The above $\left\{\left(m^{(j)}, n^{(j)}\right)\right\}_{j=0}^{\infty}$ corresponds to the solutions $\left\{\left(x_i^{(j)}, z_i^{(j)}, y_i^{(j)}, \theta^{(j)}, \mu_{i,l}^{(j)}, \kappa_{i,l}^{(j)}\right)\right\}_{j=0}^{\infty}$ of the MP^{M}. In detail, m corresponds to (x, z, y, θ) and n corresponds to (μ, κ).

Lemma B.1. If the $\mathrm{MP}_{\mathrm{SP1}}$ always has a unique solution. Then, for fixed $\rho > 0$, the inner l_1-PADM loop converges within $O(|\boldsymbol{Z}| \cdot V_{\max})$ iterations to a stationary point of MP^{M}.

Proof. Since the objective function of MP^{M} is continuously differentiable, and the $\mathrm{MP}_{\mathrm{SP2}}$ always has a unique solution. Thus, *Theorem B.1* applies. Hence, the inner l_1-PADM loop converges to a stationary point of MP^{M}.

Remark B.2. A crucial assumption of Lemma B.1 is that $\mathrm{MP}_{\mathrm{SP1}}$ always yields a unique solution. While this is a standard assumption in the literature for establishing the convergence of penalty·alternating direction methods (see [1] and [2]), we acknowledge that $\mathrm{MP}_{\mathrm{SP1}}$ (as an LP) may exhibit non-uniqueness (i.e., multiple optimal solutions). To address this, we apply an l_2-regularization term $\frac{\gamma}{2}\|q\|_2^2$ to the objective function of $\mathrm{MP}_{\mathrm{SP1}}$, where q represents the dual variables (μ, κ). This transformation converts $\mathrm{MP}_{\mathrm{SP1}}$ from an LP into a strictly convex QP, thereby guaranteeing a unique solution. Leveraging the perturbation theory established by Mangasarian [4], we note that there exists $\gamma > 0$, such that the regularized solution equals the minimum-norm solution of the original LP. In practice, we typically choose a sufficiently small γ (e.g., $\gamma = 10^{-6}$) to ensure theoretical uniqueness without introducing significant approximation error.

Next, we analyze the iteration complexity of the inner l_1-PADM loop.

The MP^{M} is rewritten in an abstract form as follows:

$$\min_{p,q}\left\{F(p,q): s.t.\ p\in\Omega^p,\ q\in\Omega^q\right\} \tag{B.1}$$

wherein $F(\cdot)$ denotes the objective function of $\mathrm{MP^M}$, $p=(x_i,z_i,y_i,\theta)$, $q=(\mu_{i,j},\kappa_{i,j})$. Then, the l_1-PADM can be summarized as two steps:

Step1 ($\mathrm{MP_{SP1}}$): Minimize $F(p,q)$ over $q\in\Omega^q$ with fixed $p=p^{(k)}$ and obtain $q^{(k+1)}$.

Step2 ($\mathrm{MP_{SP2}}$): Minimize $F(p,q)$ over $p\in\Omega^p$ with fixed $q=q^{(k+1)}$ and obtain $p^{(k+1)}$.

Trivially, $q^{(k+1)}$ and $p^{(k+1)}$ have the following properties:

$$F\left(p^{(k)},q^{(k+1)}\right)\le F(p^{(k)},q)\quad \forall q\in\Omega^q \tag{B.2}$$

$$F\left(p^{(k+1)},q^{(k+1)}\right)\le F(p,q^{(k+1)})\quad \forall p\in\Omega^p \tag{B.3}$$

By (B.3), we have

$$F\left(p^{(k+1)},q^{(k+1)}\right)\le F\left(p^{(k)},q^{(k+1)}\right), \tag{B.4}$$

By (B.2), we have

$$F\left(p^{(k)},q^{(k+1)}\right)\le F(p^{(k)},q^{(k)})\cdot \tag{B.5}$$

By (B.4) and (B.5), we have

$$F\left(p^{(k+1)},q^{(k+1)}\right)\le F\left(p^{(k)},q^{(k)}\right)\cdot \tag{B.6}$$

Thus, $F\left(p^{(k)},q^{(k)}\right)$ is a nonincreasing sequence, which also implies the optimal function value of $\mathrm{MP_{SP2}}$ is nonincreasing. Moreover, because the $\mathrm{MP_{SP2}}$ always has a unique solution, it yields that the optimal solutions of $\mathrm{MP_{SP2}}$ will not be repeated during the iteration process.

Recall that the l_1-PADM loop terminates when:

$$\left\|\left(x_i^{(j,k+1)},z_i^{(j,k+1)},y_i^{(j,k+1)},\theta^{(j,k+1)}\right)-\left(x_i^{(j,k)},z_i^{(j,k)},y_i^{(j,k)},\theta^{(j,k)}\right)\right\|\le\epsilon_1$$

Since the subproblem 2 of the l_1-PADM, i.e., $\mathrm{MP_{SP2}}$, is a MILP problem, the combinations of the binary variables are finite in number and for each combination z_i, the MILP problem turns into a reduced LP problem $LP(z_i)$ that has a polyhedral feasible region. Suppose that the solution of the $\mathrm{MP_{SP2}}$ we obtain is always the vertex of the polyhedral feasible region of $LP(z_i)$. In this situation, the number of the optima of the $\mathrm{MP_{SP2}}$ is finite.

Let $\boldsymbol{Z}$ be the collection of feasible combinations of binary variables and $V_{\max}$ be the maximum number of vertices of any of the reduced LPs. By the *pigeonhole principle* [3], the sequence must repeat a solution or meet the termination condition within $O(|\boldsymbol{Z}|\cdot V_{\max})$ iterations. By *Theorem B.1*, this solution is a stationary point of MP^M. ■

Remark B.3. In our *Lemma B.1*, we suppose that the solution of the $\mathrm{MP_{SP2}}$ we obtain is always the vertex of the polyhedral feasible region of $LP(z_i)$. It is achievable by using the branch-and-bound combined with the simplex method (also the approach adopted by Gurobi).

Lemma B.2. Suppose that $\rho^{(j)}\to\infty$, and let $\left\{\left(x_i^{(j)},z_i^{(j)},y_i^{(j)},\theta^{(j)},\mu_{i,l}^{(j)},\kappa_{i,l}^{(j)}\right)\right\}_{j=0}^{\infty}$ be a sequence of stationary points of $\mathrm{MP^M}$ (with fixed $\rho=\rho^{(j)}$) generated by the inner loop. Let $\left(x_i^*,z_i^*,y_i^*,\theta^*,\mu_{i,l}^*,\kappa_{i,l}^*\right)$ be an accumulation point of $\left(x_i^{(j)},z_i^{(j)},y_i^{(j)},\theta^{(j)},\mu_{i,l}^{(j)},\kappa_{i,l}^{(j)}\right)$ for $j\to\infty$. Then $\left(x_i^*,z_i^*,y_i^*,\theta^*,\mu_{i,l}^*,\kappa_{i,l}^*\right)$ is a stationary point of the $\mathrm{MP^M}$ with the penalty term $\sum_{l=1}^{\mathrm{k}}\sum_{i\in\mathbf{I}}\left|\Phi_{i,l}-\mu_{i,l}^T F\theta\right|=0$.

Proof. This conclusion can be directly obtained based on Lemma B.1 and Theorem B.2.
■

Lemma B.3. If there exists a finite penalty factor $\bar{\rho} \in \mathbb{R}_+$ such that for all $\rho \geq \bar{\rho}$, any convergent solution $\left(x_i^*, z_i^*, y_i^*, \theta^*, \mu_{i,l}^*, \kappa_{i,l}^*\right)$ obtained from the l_1-PADM inner loop satisfies $\Phi_{i,l} = \left(\mu_{i,l}^*\right)^\top F\theta^*, \forall i \in I, \forall l = 1, \dots, k.$, and let the ρ search method is linear search (i.e. $\rho^{(j+1)} = \alpha\rho^{(j)}, \alpha > 1$). Then the middle loop (ρ search) convergences within $O\left(log_\alpha(\bar{\rho}/\rho_0)\right)$ iterations.

Proof. The update $\rho^{(j)} = \rho_0\alpha^j \geq \bar{\rho}$ requires $j \geq log_\alpha(\bar{\rho}/\rho_0)$. Then the ρ search method convergences within $O\left(log_\alpha(\bar{\rho}/\rho_0)\right)$ iterations. ■

Remark B.4. It should be noted that in our MP^M model, we only need to use a single penalty coefficient. This is because $\sum_{l=1}^{\mathrm{k}}\sum_{i\in\mathbf{I}}\left|\Phi_{i,l} - \mu_{i,l}^T F\theta\right| = 0$ is equal to $\left|\Phi_{i,l} - \mu_{i,l}^T F\theta\right| = 0, \forall i, l$, which allows us to employ methods such as linear or binary search to quickly complete the ρ search in the middle loop. Moreover, in our *Lemma B.3*, we suppose that there exists a finite threshold of penalty factor $\bar{\rho} \in \mathbb{R}_+$. in practice, such a threshold can always be found in practice, and it is relatively small, on the order of 10^{-1} to 10^1. Moreover, since ρ only needs to be greater than this threshold to ensure that $\Phi_{i,l} = \left(\mu_{i,l}\right)^\top F\theta, \forall i \in I,\ l = 1, \dots, k.$, there is almost no need to adjust the size of ρ during the implementation of the algorithm. This also demonstrates the fault tolerance and robustness of our algorithm.

Lemma B.4. For the convergence of the MA-R&D algorithm, we have:

i) If we can get the global optimum of the $\mathrm{MP^M}$ in each iteration, then the MA-R&D algorithm converges to the actual optimal value of the SPO-UC model within $O(|\boldsymbol{Z}|)$ iterations;

ii) If $\mathrm{MP^M}$ generates a stationary solution and $\mathrm{SP_2^M}$ yields a feasible lower-level solution in each iteration, then MA-R&D converges to a feasible solution of the SPO-UC model within $O(|\boldsymbol{Z}|)$ iterations;

Proof. Note that MA-R&D reduces to the standard R&D if the global optimal value of the master problem can be obtained for each iteration. Note also that $\boldsymbol{Z}$ is finite as z is binary and R&D converges to the actual optimal value of the SPO-UC model in $O(|\boldsymbol{Z}|)$ iterations. Then, it can be seen that the first statement follows directly.

When solving the master problem heuristically to a stationary point, we obtain a feasible solution with the objective value of the master problem LB' with $LB' \geq LB$, where LB is the global optimal value of the master problem and also a valid lower bound to SPO-UC. Thus, if z^* repeats, $UB = LB \leq LB'$, which satisfies the termination condition of the MA-R&D algorithm. Moreover, if $\mathrm{SP_2^M}$ yields a feasible lower-level solution in each iteration, we can conclude that within $O(|\boldsymbol{Z}|)$ iterations MA-R&D terminates with a feasible solution to the SPO-UC model. ■

Lemma B.5. Denote that $\widetilde{\boldsymbol{Z}}$ collects the optimal values of the binary variables z obtained by $\mathrm{SP_2^M}$ in the first l iterations of MA-R&D, i.e., $\widetilde{\boldsymbol{Z}} = \{z_1^*, z_2^*, \dots, z_l^*\}$. If in the $(l+1)$th iteration, the solution of the binary variables z of $\mathrm{SP_2^M}$ belongs to $\widetilde{\boldsymbol{Z}}$, i.e., $z_{l+1}^* \in \widetilde{\boldsymbol{Z}}$, then the solution of MP in the lth iteration of MA-R&D is a stationary point of the equivalent single-level strong-duality-based reformulation of the SPO-UC ((8) in the manuscript).

Proof. Let the solution of $\mathrm{MP^M}$ defined on top of $\widetilde{\boldsymbol{Z}}$ be $\mathrm{argmin}\left(\mathrm{MP^M}(\widetilde{\boldsymbol{Z}})\right)$ $=(x^*, y^*, z^*, \theta^*, \mu_1^*, \mu_2^*, \dots, \mu_l^*, \kappa_1^*, \kappa_2^*, \dots, \kappa_l^*)$. Now, we need to prove that $\mathrm{argmin}\left(\mathrm{MP^M}(\widetilde{\boldsymbol{Z}})\right)$ (with its trivial augmented counterpart considering $\boldsymbol{Z}\backslash\widetilde{\boldsymbol{Z}}$) is a stationary point of $\mathrm{MP^M}(\boldsymbol{Z})$. Actually, if $\mathrm{argmin}\left(\mathrm{MP^M}(\widetilde{\boldsymbol{Z}})\right)$ is not a stationary point of $\mathrm{MP^M}(\boldsymbol{Z})$, then there will exist $z' \in \boldsymbol{Z}\backslash\widetilde{\boldsymbol{Z}}$ rendering some constraint(s) of $\mathrm{MP^M}(\boldsymbol{Z})$ fail to hold. In this situation, the solution z of the $\mathrm{SP_2^M}$ in the $(l+1)$th iteration will be an element of $\boldsymbol{Z}\backslash\widetilde{\boldsymbol{Z}}$, i.e., $z_{l+1}^* \in \boldsymbol{Z}\backslash\widetilde{\boldsymbol{Z}}$, which contradicts with

$z_{l+1}^{*} \in \widetilde{\boldsymbol{Z}}$. This completes the proof. ■

The next result follows directly from *Theorem B.1*, which ensures the convergence of MA-R&D under some mild conditions.

Proposition B.1 (Proposition 2 in the manuscript)**.** The MA-R&D algorithm converges to a stationary point of the equivalent single-level strong-duality-based reformulation of the SPO-UC model within $O(|\boldsymbol{Z}|)$ iterations as $\rho^{(j)} \to \infty$, if the $\mathrm{MP}_{\mathrm{SP1}}$ always has a unique solution and $\mathrm{SP}_2^{\mathrm{M}}$ yields a feasible lower-level solution in each iteration.

Proof. This conclusion can be obtained based on *Lemma B.4* ii) and *Lemma B.5*. ■

Remark B.5. It should be noted that the iteration complexity of $O(|\boldsymbol{Z}|)$ might not be trivial. However, as shown in Section V, the algorithm only requires a small number of iterations to converge, which is significantly less than $|\boldsymbol{Z}|$.

Reference

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